# SHREYANSH RAI

**Full Stack Developer | MERN Stack Specialist | AI & ML Engineer**

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## PROFESSIONAL PROFILE

Ambitious Full Stack Developer and AI & ML Engineer with 1+ year of hands-on professional experience building scalable MERN Stack applications and advanced deep learning systems. Co-built and deployed accudesign.in a fully functional, production-grade service platform at Technewity Labs, Pune. As lead researcher and author, architected a Hybrid Deep Learning fraud detection system (BiLSTM + XGBoost + Autoencoder) published at the ACROSET 2026 IEEE conference, achieving 99.9% recall on 6.36M UPI transaction records. Adept at bridging full-stack engineering and AI/ML research, with strong command over the entire web development lifecycle, from MongoDB schema architecture and FastAPI backend design to React.js component development and cloud deployment.

## TECHNICAL SKILLS

**Frontend:** React.js, HTML5, CSS3, JavaScript (ES6+), React Hooks, Context API, Responsive Web Design, DOM Manipulation, JSX
**Backend:** Node.js, Express.js, FastAPI (Python), RESTful API Design, JWT Authentication, OAuth 2.0, Middleware, MVC Architecture, Session Management, Helmet.js, CORS
**AI / ML & Deep Learning:** TensorFlow/Keras, Scikit-learn, XGBoost, Random Forest, BiLSTM, Bahdanau Attention, Autoencoders, Isolation Forest, SMOTE, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Streamlit, Joblib, ROC-AUC
**Database:** MongoDB, Mongoose ODM, Schema Design, Aggregation Pipelines, Compound Indexing, Query Optimization, Field-level Validation, Repository Pattern
**DevOps / Tools:** Git/GitHub, Postman, Vercel, Render, VS Code, Jupyter
**Emerging Tech:** IoT Prototyping (Arduino IDE, ESP32, DHT11 Sensors), AI Chatbot API Integration

## WORK EXPERIENCE

**Full Stack Developer Intern** · Technewity Labs, Pune *Jan 2025 – Apr 2026*

- Led frontend development of ACCU DESIGN (accudesign.in) using React.js with functional components, React Hooks, and modular Tailwind CSS fully responsive, production-deployed UI serving real enterprise clients.
- Designed and implemented a secure authentication system using JWT tokens and role-based access control (RBAC), supporting distinct Admin and User permission tiers across all API routes.
- Built 15+ RESTful API endpoints in Express.js with middleware-based validation, error handling, and structured JSON response patterns; optimized MongoDB schemas for 40%+ query performance gain.
- Integrated AI chatbot API for intelligent user interaction and applied AI/ML research insights into web application architecture, bridging full-stack and AI engineering workflows.
- Deployed to production using Render (frontend and backend) with environment variable management and Git version control via feature branching and Agile sprints.

**Web Development Intern** · Codespaze, Mumbai *Jul 2024 – Sep 2024*

- Developed and enhanced frontend web components across client projects using HTML5, CSS3, JavaScript; improved page load performance through code optimization.
- Assisted in REST API integration on backend tasks; maintained project repositories using Git branching strategies and participated in Agile sprint reviews and daily standups.

## RESEARCH PUBLICATION

**Precise and Enhanced Identification of UPI-based Transaction Scams using Hybrid Ensemble Model**

*Lead Author | Published at ACROSET 2026 IEEE Conference | Paper ID: 360*

- Designed the full V5 Hybrid Deep Learning architecture: a dual-path evaluation engine combining supervised classifiers (XGBoost + Random Forest) for instant blocking of known fraud patterns, with an unsupervised deep learning tier (BiLSTM + Bahdanau Attention + Autoencoder + Isolation Forest) for zero-day novel fraud detection.
- Engineered 19 behavioural velocity features from the PaySim dataset (6.36M records); applied SMOTE oversampling and Z-score normalization to handle extreme class imbalance (0.13% fraud ratio), achieving 99.9% Recall, 99.8% Precision, and 99.9% AUC-ROC.
- Built a high-performance asynchronous FastAPI backend capable of processing 10,000+ real-time transaction evaluations per second with sub-50ms latency, bypassing standard database overhead.

- Deployed an interactive Streamlit dashboard with live transaction monitoring, explainable AI fraud alerts, dynamic blocking rule enforcement, and a built-in zero-day attack simulator for security analysts.

## PROJECTS

**ACCU DESIGN — Industry-Level MERN Service Platform live at [accudesign.in](accudesign.in)** *Jan 2024 – Present*

*Tech Stack: React.js · Node.js · Express.js · MongoDB · JWT · OAuth · Helmet.js · Git · Render ·*

- Separated concerns across distinct service layers (Auth Service, Order Service, User Service) following a layered MVC pattern — Route → Middleware → Controller → Model — laying the groundwork for future microservices migration and keeping business logic fully decoupled from API handlers.
- Applied CORS policy configuration restricting cross-origin requests to whitelisted domains only, preventing unauthorized API consumption; integrated Helmet.js to enforce secure HTTP headers (CSP, X-Frame-Options, HSTS), hardening the Express server against common web vulnerabilities.
- Designed MongoDB schemas with explicit field-level validation, enum constraints, default values, and auto-managed timestamps (createdAt, updatedAt) ensuring data integrity at the persistence layer and eliminating invalid state from entering the database.
- Implemented a centralized Axios instance on the frontend with request/response interceptors handling JWT token injection, expiry detection, and automatic redirect to login — eliminating repetitive auth logic across all 15+ API call sites.
- Built role-based access control (Admin/User/Vendor) with protected routes and permission-gated API access; admin panel includes user management, order status updates, analytics dashboard, and service configuration.

**UPI Fraud Detection System — Hybrid Deep Learning Research** *2025*

*Tech Stack: Python · TensorFlow/Keras · XGBoost · Random Forest · BiLSTM · Autoencoder · Isolation Forest · FastAPI · Streamlit · Pandas · NumPy · SMOTE · Scikit-learn*

- Architected a dual-path hybrid AI engine: Path A (XGBoost + Random Forest) auto-blocks known fraud signatures using 19 engineered features with near-zero latency; Path B (BiLSTM + Bahdanau Attention + Autoencoder + Isolation Forest) detects novel zero-day attacks via temporal sequence analysis and reconstruction-error anomaly scoring.
- Built a scalable asynchronous FastAPI backend with a custom Python parser bypassing standard database overhead, sustaining 10,000+ transaction ingestion requests per second at sub-50ms latency designed for production-grade high-frequency financial systems.
- Applied a strict stratified 70/15/15 train/validation/test split preserving the realistic 0.13% fraud ratio; used SMOTE exclusively on the training set to augment fraud samples to ~10% without leaking distribution information into evaluation sets.
- Implemented weighted ensemble voting with adaptive 0.77 auto-thresholding and five-step temporal escalation logic across four specialized models — outperforming all standalone baselines: 99.7% accuracy vs Gradient Boosting (98.4%), Random Forest (94%), F1-Score of 99.88%.
- Integrated end-to-end ML pipeline: data ingestion → PaySim feature engineering → Z-score normalization → dual-path model inference → three-tier classification (BLOCK / BLOCK_NOVEL / REVIEW / ALLOW) → Streamlit dashboard with explainable AI alerts and live fraud blocking rules.

**Library Management System — Role-Based Digital Platform** *2024*

*Tech Stack: React.js · Node.js · Express.js · MongoDB · JWT · Mongoose · Git*

- Designed MongoDB schema for a many-to-many book-user borrowing relationship using Mongoose referencing with explicit field-level validation on all required fields and auto-managed timestamps, ensuring referential integrity across borrow and return lifecycle events.
- Implemented JWT-secured login with protected API routes enforcing role-based middleware , admin-only operations (add/remove books, manage users, view borrowing history) are rejected at the middleware layer before reaching any controller logic.
- Structured backend with a clean separation between route definitions, auth middleware, and controller logic, each concern isolated to its own module, making the codebase extensible for additional roles or borrowing rule changes without modifying existing handlers.

**Blogs Management Website — Full Stack Publishing Platform** *2024*

*Tech Stack: React.js · Node.js · Express.js · MongoDB · REST APIs · Git*

- Implemented an admin approval workflow as a stateful lifecycle on the post schema — posts transition through PENDING → APPROVED/REJECTED states stored in MongoDB, with API routes enforcing that only admin-role tokens can trigger state transitions.
- Applied CORS configuration to restrict API access to the frontend origin only; used schema-level enum validation on post status fields to prevent invalid state writes directly at the database boundary.
- Built fully responsive UI with CSS Grid and Flexbox adapting across mobile, tablet, and desktop viewports; used JWT auth with role management for user self-registration and admin-only content moderation routes.

**IoT-Based Smart Weather Station & Irrigation System** *2023*

*Tech Stack: Arduino IDE · ESP32 · DHT11/22 Sensors · Rainfall Sensor · C++ · WiFi*

- Programmed sensor reading logic and decision-making algorithms in C++ on the ESP32 microcontroller — temperature, humidity, and rainfall sensor data are sampled on configurable intervals, compared against irrigation threshold constants, and used to autonomously trigger relay-controlled irrigation without manual intervention.

- Transmitted real-time environmental data over WiFi using the ESP32's built-in network stack, enabling remote monitoring of sensor readings; system architecture separates sensor acquisition, threshold evaluation, and actuation into distinct functional blocks for maintainability.

**AI-Based BMI Recommendation System** *2023*

*Tech Stack: HTML5 · CSS3 · JavaScript · DOM Manipulation · AI Chatbot API*

- Integrated an AI chatbot API with a stateless request pattern — each user query sends the current BMI classification and user profile as context in the API payload, enabling the model to generate personalized, context-aware diet and exercise recommendations without maintaining server-side session state.
- Built entirely in Vanilla JS with no framework dependency — DOM manipulation, input validation, BMI classification logic, and API call handling all implemented using native browser APIs, demonstrating strong foundational JavaScript and separation of UI logic from data processing.

## EDUCATION

**B.E. Computer Science (AI & ML) — ARMIET, Mumbai University** *2022 – 2026 | CGPA: 8.63 / 10*

HSC (12th Grade) — ISC Board, St. Joseph Montessori School, Lucknow *2021 | 89.27%*

SSC (10th Grade) — ICSE Board, St. Joseph Montessori School, Lucknow *2019 | 83%*

## CERTIFICATIONS & ACHIEVEMENTS

- Full Stack Web Development Internship Certificate — Technewity Labs, Mumbai (2025)
- Python 3.X Programming — Simplilearn SkillUp (Feb 2024, Code: 4883339)
- MumbaiHacks 2025 — Largest Agentic AI Hackathon participant, organized by TEAM Mumbai & Made in Mumbai (Nov 2025)
- Tata Crucible Campus Quiz — National-level Business & GK Prelims qualifier (Nov 2025)
- HTML Certification by Complete Coding
- CSS Certification by Complete Coding

## LANGUAGES & INTERESTS

**Languages:** Hindi (Native) | English (Professional / Intermediate)

**Interests:** AI/ML Research · Open Source Contribution (MERN/React) · IoT & Embedded Systems · System Design & Scalable Architectures